EX- 2: 8 Queen problem

Program:

```
N = 8
board = [-1] * N

def is_safe(row, col):
    for i in range(row):
        if board[i] == col:
            return False
        if abs(board[i] - col) == abs(i - row):
            return False
    return True

def solve(row):
    if row == N:
        return True

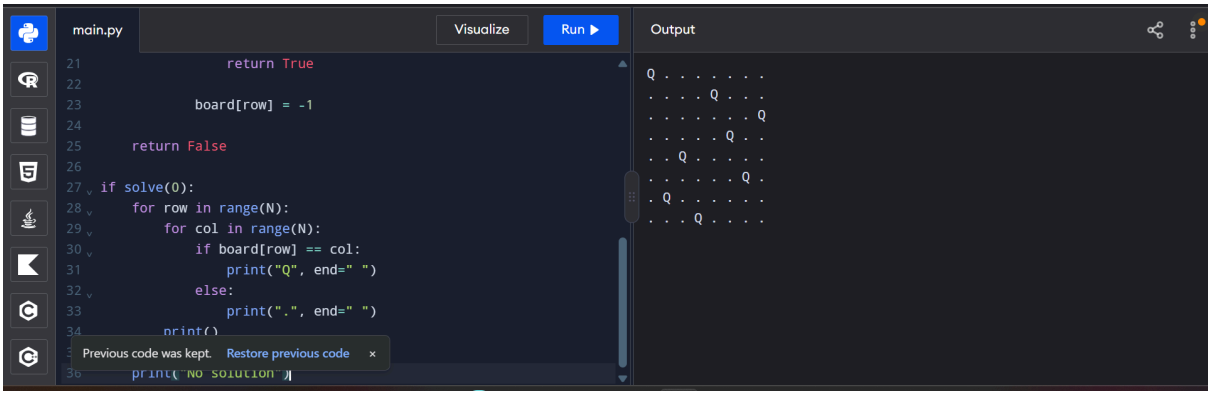
    for col in range(N):
        if is_safe(row, col):
            board[row] = col

            if solve(row + 1):
                return True

            board[row] = -1

    return False

if solve(0):
    for row in range(N):
        for col in range(N):
            if board[row] == col:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()
else:
    print("No solution")
```


main.py
Visualize
Run
Output
return True
board[row] = -1
return False
if solve(0):
for row in range(N):
for col in range(N):
if board[row] == col:
print("Q", end=" ")
else:
print(".", end=" ")
print()
Previous code was kept.
Restore previous code
print("No solution")
Q . . . . . .
. . . . Q . . .
. . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .